

## 1. Bill of Materials (Final)

The following table lists all components used in the Smart Home Automation System, verified against the final built project.

Prices are approximate PKR market rates (ELECTROBES. Rawalpindi/Islamabad).

| #  | Component                | Designator | Specifications                           | Qty   | Unit Price (PKR) | Total (PKR)  | Actual Component Used |
|----|--------------------------|------------|------------------------------------------|-------|------------------|--------------|-----------------------|
| 1  | ESP32-S3 Microcontroller | ARD1       | Dual-core 240MHz, Wi-Fi + Bluetooth      | 1     | 1,600            | 1,600        | ESP32-S3              |
| 2  | Ultrasonic Sensor        | SONAR1     | HC-SR04, Range: 2-400 cm, 5V             | 1     | 200              | 200          | HC-SR04               |
| 3  | MQ-2 Gas Sensor          | GAS1       | Detects LPG, Smoke, CO, Analog output    | 1     | 300              | 300          | MQ-2                  |
| 4  | Servo Motor (180 deg)    | MOT1       | SG90, 180 degree rotation, 5V            | 1     | 300              | 300          | SG90 Servo            |
| 5  | DC Motor (Water Pump)    | MOT2       | Mini submersible pump, 3-6V DC           | 1     | 200              | 200          | Mini DC Pump          |
| 6  | Buzzer                   | BUZ1       | Active buzzer, 5V                        | 1     | 50               | 50           | Active Buzzer         |
| 7  | LED (Red)                | D1         | 5mm Red LED, 2V forward voltage          | 1     | 10               | 10           | 5mm Red LED           |
| 8  | Resistor                 | R1, R2, R3 | 100 Ohm, 1/4W                            | 3     | 5                | 15           | 100 Ohm               |
| 9  | Potentiometer            | RV1, RV2   | 1k Ohm trimmer potentiometer             | 2     | 30               | 60           | 1k Trimmer            |
| 10 | LCD Display (I2C)        | LCD1       | 16x2 LCD with I2C module, 5V             | 1     | 350              | 350          | 16x2 I2C LCD          |
| 11 | Breadboard               | -          | 830-point solderless breadboard          | 1     | 150              | 150          | Breadboard            |
| 12 | Veroboard                | -          | General purpose PCB stripboard           | 1     | 100              | 100          | Veroboard             |
| 13 | Soldering Wire           | -          | 60/40 rosin core solder wire             | 1     | 100              | 100          | Solder Wire           |
| 14 | Header Pins              | -          | Male/Female 2.54mm header pins           | 1 set | 50               | 50           | Header Pins           |
| 15 | Buzzer + LED + Wires     | BUZ2, D2   | Passive buzzer, LED, jumper wires bundle | 1 set | 150              | 150          | Bundle                |
|    |                          |            |                                          |       | <b>TOTAL</b>     | <b>3,235</b> |                       |